The triangle inequality

1 For the following:

i Complete the statements, using $=$, $>$, or $<$.

ii State whether or not the following can be sides of a valid triangle.

a 7, 12 and 17.

- $7 + 12 \square 17$
- $7 + 17 \square 12$
- $17 + 12 \square 7$

b 3.2, 7.4 and 9.6.

- $3.2 + 7.4 \square 9.6$
- $3.2 + 9.6 \square 7.4$
- $9.6 + 7.4 \square 3.2$

c 9, 4 and 5.

- $9 + 4 \square 5$
- $9 + 5 \square 4$
- $5 + 4 \square 9$

d 3.1, 7.7 and 11.8.

- $3.1 + 7.7 \square 11.8$
- $3.1 + 11.8 \square 7.7$
- $11.8 + 7.7 \square 3.1$

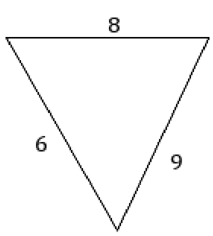
2 Consider three sides 8, 8 and 8 :

a Complete the statement $8 + 8 \square 8$, using $=$, $>$, or $<$:

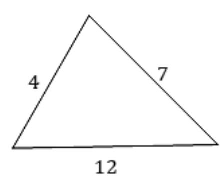
b State whether or not the three sides can make a valid triangle.

3 State whether or not the following triangles are valid.

a



b



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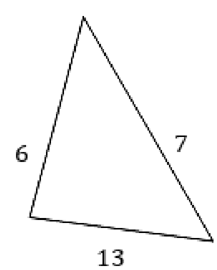
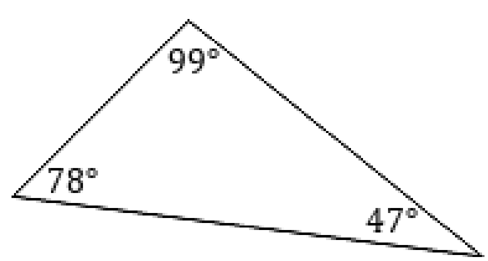
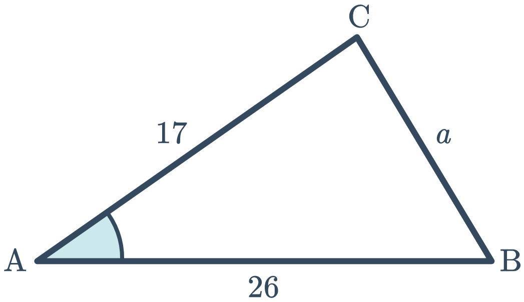


4 Why is it not possible to make the following triangle?

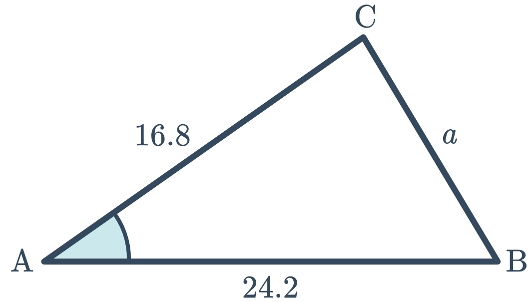


5 For the following diagrams, determine the range of values for a.

a



b



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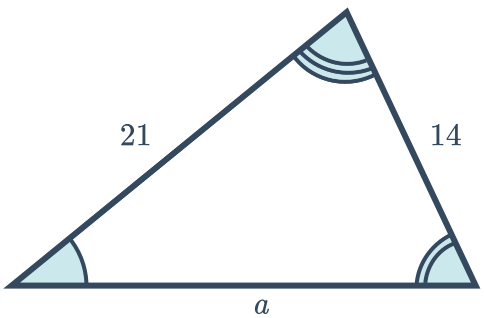
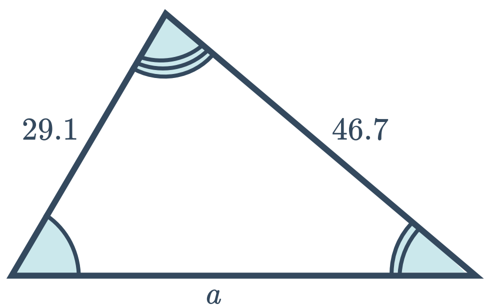
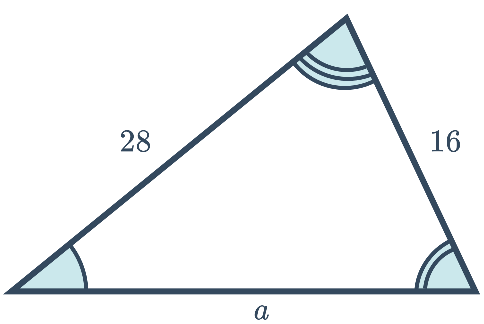


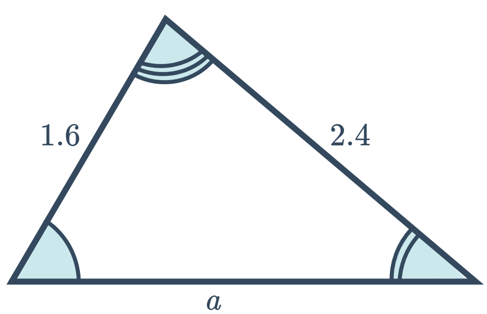
d



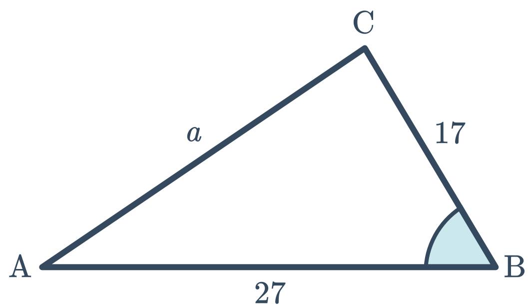
e

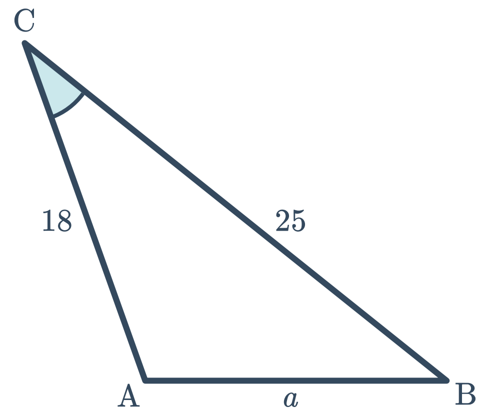


f



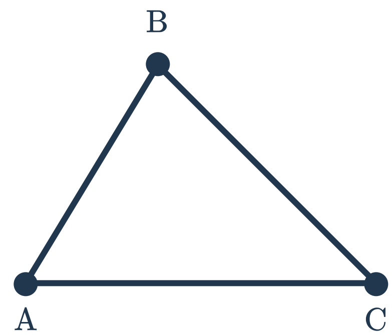
g





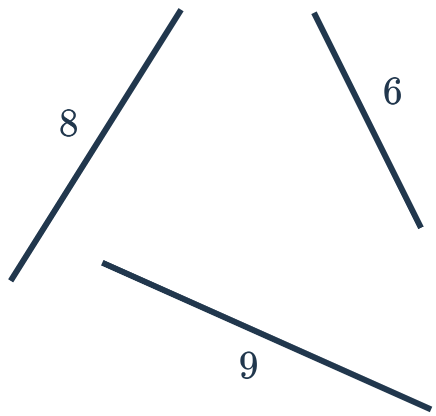
Additional questions

6 For the following triangle, state the triangle inequality theorem in words and using notation.

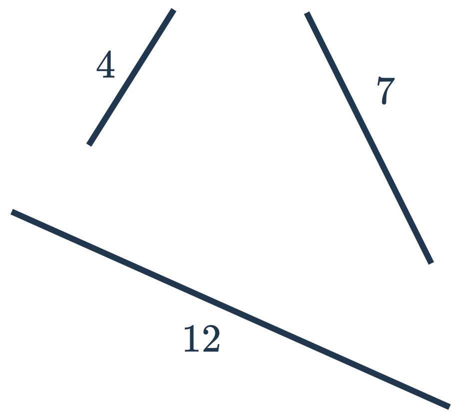


7 State whether or not the following lengths can make a valid triangle. If it is not valid, state which inequality fails.

a



b



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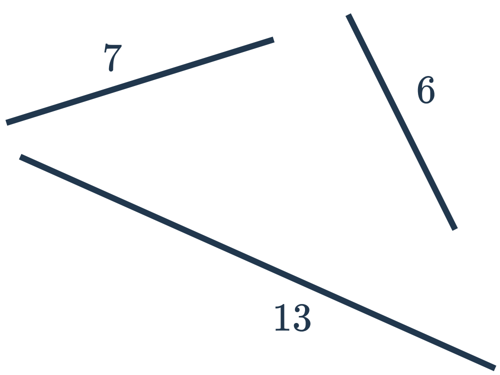
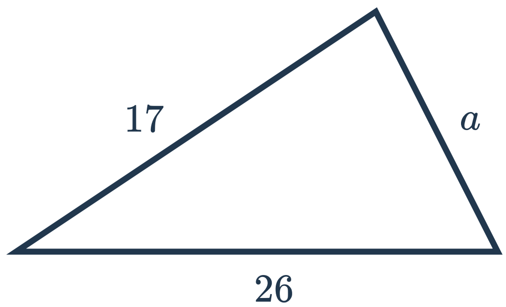


8 For each of the given triangles:

i State and simplify the three triangle inequalities

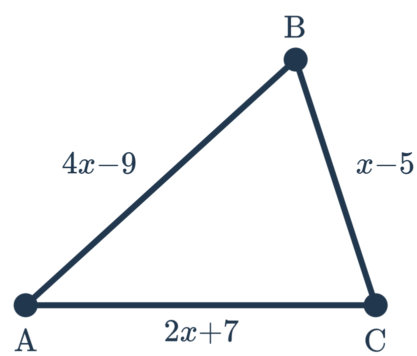
ii Determine the range of possible values for a

a

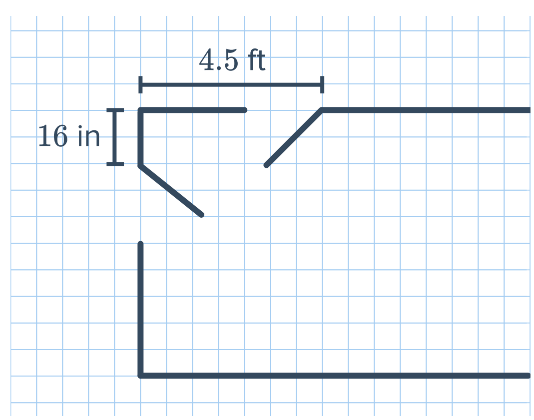


b $\triangle ABC$ with side lengths of 15, 18, and a

9 Determine the possible range of values for x using the given figure.



10 A contractor is creating the plans for a house and a hallways has two standard doors very close by. He wants to have all of the doors open in to the hallways, as seen in the given diagram. A standard door has a width of 36 inches.
The contractor has approved the plans and said that the doors will never hit each other because the triangle inequality says there is no valid triangle using those lengths.
Identify and explain his error.



11 When leaving Jacksonville and travelling along I-95, Aurelia sees the given road sign and says, "Wow, this means that it is only 204 miles from Orlanda to Miami." Determine if this is a valid conclusion. If so, justify it. If not, explain her error.